METRIC
A-A-30194
July 26, 1993

COMMERCIAL ITEM DESCRIPTION

LINER, GRAVE, CONCRETE

The General Services Administration has authorized the use of this commercial item description as a replacement for VA Specification X1708G for concrete.

This commercial item description describes Concrete Graveliners intended for use in national cemeteries to prevent the sinkage of soil in graves.

Salient characteristics. LINER, GRAVE, CONCRETE - Shall be constructed from reinforced concrete utilizing Portland Cement. Cement will be of commercial quality using steel and/or polypropylene reinforcement. In addition the material(s) used in the production of concrete graveliners shall be impervious to any destructive action of the natural elements possibly contained in the cemetery soil which can range from extreme acidic to extreme alkaline conditions.

Concrete graveliners must be properly cured for 28 days before delivery and the concrete must have a minimum compression strength of 31,027 kilopascal (kPa) or 4,500 pound-force/square inch (PSI).

Style 1 - Box with separate lid
Style 2 - Dome with a base
Style 3 - Dome without a base

SIZES	INTERNAL DIMENSIONS
a. Small Liner	147 cm (58 in) by 51 cm (20 in) by 46 cm (18 in)
b. Medium Liner	218 cm (86 in) by 76 cm (30 in) by 61 cm (24 in)
c. Oversize Liner	229 cm (90 in) by 81 cm (32 in) by 66 cm (26 in)
d. Extra Large Liner	234 cm (92 in) by 91 cm (36 in) by 71 cm (28 in)

Internal dimensions listed above are nominal for the standard sizes and small variances are permitted in the use of standard commercial products providing size permits use of standard caskets.

FSC 9930

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UNIT: EACH

Workmanship. Graveliners shall be free from defects which detract from its appearance or impair its serviceability.

Production specification. The following must be included as part of the bid:

a. Type of material or materials
b. Material specifications
c. Methods of construction
d. Testing procedures and findings

Performance. Graveliners furnished will be designed for a maximum burial depth of 2.13 m (84 in) from the bottom of the liner, with minimum soil cover of 46 cm (18 in), be capable of structurally withstanding passage of an axle load of 5.0 metric tons (11,000 lbs.) after burial and provide for ease of installation. Liners will have a sufficient weight bearing surface area to prevent sinkage. This area may vary depending upon soil conditions, but in all cases, will be capable of passing the functional test specified in paragraphs entitled "Functional tests" in order to perform this function in place after installation. The ongoing static loads in this environment include the weight of the backfill, changes in the water table, and pressures from the freeze-thaw cycle where applicable. In addition, the graveliner shall be designed to withstand the dynamic, differential and impact loads and stresses which are exerted upon a unit during the interment process. Such loads and stresses include compaction of the (backfill) soil, excavation of adjoining gravesite, as well as the pressures and forces exerted by normal cemetery excavation, compaction and maintenance equipment. Graveliner designs shall be such as to prevent any potential sinkage due to chiseling (The sinkage process caused because of insufficient surface area contact with the bottom of the grave.) and fatigue creep (The long run sagging process of the weight bearing surface area.).

Design and construction. Graveliners will meet the performance requirements. The combination of the design shape, design thickness, structure and material shall be capable of supporting the required 2.27 metric tons (5,000 lbs.) center load pressure test and the axle load test of 5.0 metric tons (11,000 lbs.), plus the superimposed earth load. Both the exterior dimensions and the weight of the graveliner shall be kept to a minimum in accordance with the normal internal dimension standards specified above in Paragraph labeled "Salient characteristics" and the required structural strength specified above. Design contours and shapes will be such as to eliminate any possible back-fill voids, settling, sinking, floating or raising of the graveliner once

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interred. The graveliners and parts thereof, will have provisions to facilitate both the ease of installation and removal of the unit and its component parts. All lifting apparatus must be adequately anchored in the unit to sufficiently withstand the stresses of lifting and placing the unit. Graveliners will have knockout holes to permit drainage.

Inscription. The date of manufacture will be permanently recorded on the graveliner so that the contractor can identify his product according to the date of production.

QUALITY CONTROL AND ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Preproduction sample. When specified a preproduction sample will be required and will be examined and tested in accordance with this specification. The preproduction sample will be representative of the production article. However, the requirements for furnishing a preproduction sample may be waived as to a bidder if:

a. The bidder states in the bid that the product offered is the same as a product previously furnished to the procuring activity under a prior contract, and the bidder identifies the contract.

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b. The contracting officer determines that such product meets the requirements of the specification.

Tests. Tests shall be conducted to determine compliance with these commercial item description requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Testing of components. Upon request, the supplier will furnish a certification of compliance, issued by an independent testing laboratory or an approved testing program. This would include an analysis for all characteristics of material specified and applicable referenced specifications. The Government reserves the right to verify the correctness of such certificates of compliance and analysis. Approved graveliners are subject to both random compliance and complete re-testing.

The graveliners will conform with the following American Society for Testing and Materials (ASTM) Standards.

ASTM C 33 - Standard Specifications for Concrete Aggregate
ASTM C 39 - Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C 40 - Standard Test Method for Organic Impurities in Fine Aggregates for Cement
ASTM C 94 - Standard Specification for Ready-Mixed Concrete
ASTM C 150 - Standard Specification for Portland
ASTM C 173 - Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method
ASTM C 231 - Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C 260 - Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C 330 - Standard Specification for lightweight Aggregates for Structural Concrete
ASTM C 494 - Standard Specification for Chemical Admixtures for Concrete

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-1.

Examination of furnished product. Graveliners will be examined for the defects listed below. Whenever a deviation to specified requirements is noted, correction will be made. Failure to make corrections will be cause for rejection:

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a. Size/dimensions not as specified.
b. Materials not as specified.
c. Production date not inscribed or otherwise permanently recorded.
d. Drainage holes not provided.
e. Surface contains foreign inclusions, spalling, holes, cracks, large voids, or other evidence of poor manufacture.
f. Provisions to facilitate installation and removal not provided.

Functional Test. Upon request, a functional load test will be made at the contractor's expense under the direction of the contracting officer, or his/her representative, to insure the graveliner, as furnished, will be capable of supporting load weight stated in paragraph entitled "Design and construction". The functional test will consist of two loading conditions:

a. Unconfined loading. (Center Load Pressure Test)

1. Place base on 10.16 cm (4 in) x 10.16 cm (4 in), spaced approximately 0.305 m (12 in), from each end of the test product.

2. Place cover on base. The producer may choose to use a temporary gasket material in the sealant (tongue and groove) area of a graveliner in place of the permanent sealant.

3. A test frame will be assembled around the center of the test unit.

4. Cushion material should be used to assure that a 0.093 sq. m (1 sqft), bearing plate will uniformly load the test unit. Special care should be taken for units with arch-shaped tops or carapaces.

5. A hydraulic ram will be pressurized until the ram engages the bearing place, and will be set at a preload pressure of 226.8 kg (500 lb) for a minimum of one minute.

6. The inspector will add additional pressure to the test unit in 2,268 kg (5,000 lb) increments. This pressure will be added gradually, taking at least thirty (30) seconds to add each additional 226.8 kg (500 lb) load. Once the 226.8 kg (500 lb) increment has been added, the test unit should be allowed a minimum of one minute at rest prior to the next load increase. This procedure should be repeated until the full 2,268 kg (5,000 lb) load has been properly applied.

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7. The concrete graveliner must withstand the 2,268 kg (5,000 lb) of static load on the 0.93 m (36 in), section at the critical load area for a minimum of five (5) minutes without failure.

b. Confined loading.

1. The liner shall be placed on a flat surface below ground and covered with a granular soil material. The soil will be compacted thoroughly around and on top of the liner to a thickness of 46 cm (18 in). A wheeled tractor or truck with an axle load of 5.0 metric tons (11,000 lbs.) will then be passed over the covered liner for a minimum of twenty (20) times in repetition. The liner must not show any signs of material overstress or cracking.

2. Upon completion of Confined loading test the adjacent "gravesite" will then be excavated to a maximum depth of 2.13 m (84 in), exposing the longitudinal side of the tested unit for a period of no less than twenty-four (24) hours. The unit must not exhibit any signs of potential material fatigue, creep, chiseling, excessive deflection, scratching or infringement of the adjacent gravesite.

The contracting officer or his/her designated representative at any time may make unannounced inspections of the contractor's plant.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specification, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act - If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal *Food, Drug and Cosmetic Act, as amended, and regulations* promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its contractors/suppliers, with the requirements of all other applicable Federal, State, and local statues, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Packaging and packing. Graveliners will be palleted or otherwise prepared for shipping in accordance with standard commercial practice to insure carrier acceptance and safe delivery in containers complying with rules and regulations applicable to the mode of transportation.

Marking. Shipping containers will be marked in accordance with the manufacturer's standard commercial practice. The production date will be inscribed or otherwise permanently marked on each component which makes up a graveliner.

Copies of Federal Standard No. 376, Preferred Metric Units for General Use by the Federal Government and Federal Test Standard 191 are available from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

Copies of Specifications and Standards are also available from Department of Defense Single Stock Point. Special Assistance Desk, telephone (215) 697-2667 or 2179, will provide you adequate information by mail or TeleSpecs mode.

Copies of United States Pharmacopoeia (USP) regulations may be obtained from: USPC, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852, USA.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description
b. Type, style and size.

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MILITARY INTERESTS: CIVIL AGENCY COORDINATING ACTIVITIES

Military Coordinating Activity: FDA-MPQAS
PHS

DoD-MD

PREPARING ACTIVITY:

VA-OSS VANAC (904)
Agent VA-OSS VACO (92A):

AGENT: DLA-DM

Custodians:

Army - MD
Navy - MS
Air Force - 03

Project No. 9930-2395

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